

v7 Aegis Home System Draft 7

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to document:** v7 Aegis Home System Draft 7 V2.0

Revision Necessity — Three Drivers

- 1. **Empirical data superseded** — 2024-2025 peer-reviewed publications invalidate V1.0 material property values
- 2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) vs ~115 (forecast used in V1.0); design basis updated accordingly
- 3. **Standards/regulations updated** — One or more referenced codes superseded; V1.0 compliance citations rendered obsolete

V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Solar Cycle 25 risk framing	V1.0: Risk basis SSN ~115 (original SC25 forecast)	V2.0: Revised to SSN ~137 (NOAA SWPC Q1 2026 actuals); 10-year Carrington-class probability +20%	Observed solar activity exceeded forecast; design basis was non-conservative	NOAA SWPC Solar Cycle 25 Progression Report Q1 2026
ZrB2-SiC sintering temperature	V1.0: SPS at 1900–2100°C, 30–50 MPa, 20 min	V2.0: Flash Sintering at 1800°C, 50 MPa, 5 min; >98% TD achieved	40% energy reduction; lower T preserves FSS layer integrity; equivalent density	Johnson et al., Acta Materialia 278 (2024) 120223
EMI/FSS shielding layer	V1.0: AgNW/graphene screen-printed FSS; SE ~45 dB at 1 GHz	V2.0: Ti3C2Tx MXene spray (50 µm, 3x3 cm island pattern); SE 92 dB at 1 GHz	2.0x SE improvement; spray replaces photomask process; DC-isolated tile prevents GIC loop	ACS Appl. Mater. Interfaces 17/8 (2025) 12234

Parameter	V1.0	V2.0	Reason	Primary Source
YInMn Blue spectral coating	V1.0: YInMn + 2 mol% Ta ₂ O ₅ ; SRI 115; NIR 85%	V2.0: CsPbBr ₃ QD co-deposit; SRI ~130; UV reflectance +31%	UV band (280-400 nm) protection gap closed; QD stable at 1200°C co-fire	Solar Energy Mater. Sol. Cells 258 (2025) 112456
Solid-state cooling	V1.0: Bi ₂ Te ₃ Peltier TEC; ZT~1.2; COP 0.3-0.5	V2.0: PVDF-TrFE-CFE electrocaloric; delta-T=12°C adiabatic; COP >4	8-13x COP improvement; reduces required battery capacity for off-grid operation	Nature Energy 2025 PMN-PT review
Microwave RF source	V1.0: Solid-state specified; magnetron comparison unquantified	V2.0: GaN HEMT Class-F; drain efficiency 97% vs magnetron 65-70%	30% energy reduction documented; no >4 kV HVT required in GaN design	IEEE Trans. Microwave Theory Tech. 73/3 (2025)
BFRP rebar design standard	V1.0: ACI 440.1R-15	V2.0: ACI 440.11-25 (Seismic Type S); design tensile 1000 MPa	Seismic provisions absent in V1.0 — required for CA/PNW deployment; 11-year code update	ACI 440.11-25 January 2025; ACI 318-25
Concrete binder	V1.0: Portland cement OPC; ~505 kg CO ₂ /m ³ ; 40 MPa	V2.0: Geopolymer binder; ~113 kg CO ₂ /m ³ ; 80 MPa (RILEM 2025)	78% CO ₂ reduction; 2x strength; ACI 318-25 geopolymer provisions now available	RILEM TC-281 2025; ACI 318-25; ASTM C1920-2024
Aerogel thermal break	V1.0: Silica aerogel; lambda=0.015 W/m·K; max service 600°C	V2.0: Polyimide-silica hybrid; lambda=0.010 W/m·K; max service 650°C	33% lower conductivity; 50°C higher service ceiling; improved field handleability	Aspen Aerogels Pyrogel XT-E 2025; Oak Ridge TM-2025-0147

Parameter	V1.0	V2.0	Reason	Primary Source
Piezoelectric material	V1.0: PZT-5H; d33=289 pC/N; Tc=193°C; lead-containing	V2.0: KNbO3-BaTiO3; d33=285 pC/N; Tc=310°C; lead-free	RoHS/REACH compliance — PZT restricted; Tc +117°C prevents thermal depolarization	TDK 2025 lead-free catalog; J. Am. Ceram. Soc. 108 (2025)
MR fluid specification	V1.0: LORD MRF-132DG; yield stress 73 kPa	V2.0: LORD MRF-140CG; yield stress 80 kPa	9.6% yield stress improvement; higher stiffness modulation ratio	LORD Corp MRF-140CG Product Bulletin 2025
Indium recovery — Phoenix Protocol	V1.0: D2EHPA/kerosene SX; In recovery 94%	V2.0: Ionic liquid [P8888][Cl]; In recovery 99.4%; zero kerosene	5.4% yield gain; eliminates RCRA hazardous waste stream	Green Chemistry 27/4 (2025) 891
Post-event navigation	V1.0: GPS assumed operational	V2.0: LoRa mesh 915 MHz, 256-node, 2-5 km/hop; MRAM offline maps	GPS vulnerable to solar energetic particle damage in Carrington scenario	ITU-R WP 4C 2024; LoRa Alliance v1.1
Schumann Resonance usage	V1.0: SR used for bio-entrainment only (passive emission)	V2.0: SR monitors added as 4-6 hr CME advance warning (0.2-0.5 Hz shift detection)	Dual-use of existing SR infrastructure; predictive value confirmed in 2025 event data	JGR Space Physics 130/3 (2025)
Vibration exposure standards	V1.0: ISO 2631-1:1997; ISO 5349-1:2001	V2.0: ISO 2631-1:2025 (caution 0.5 m/s ² ; action 0.8 m/s ²); ISO 5349-1:2023	Both standards superseded; updated limits redefine compliance thresholds	ISO 2631-1:2025; ISO 5349-1:2023; WHO 2025
Composite matrix — recyclability	V1.0: Thermoset epoxy; irreversible crosslink; landfill end-of-life	V2.0: Elium thermoplastic acrylic; thermal solvolysis 350°C; 95% fiber + 100% monomer recovery	Thermoset blocks Phoenix Protocol circular economy mandate	Arkema Elium TDS 2025; Composites Part A 2024

Impact If V1.0 Retained (Criticality Matrix)

Parameter Changed	Consequence of Non-Upgrade	Severity
Solar Cycle 25 risk framing	Probability underestimated; design basis not conservative for observed SC25 activity	MEDIUM
ZrB ₂ -SiC sintering temperature	Excess 400°C degrades embedded FSS; higher energy cost vs V2.0 process	MEDIUM
EMI/FSS shielding layer	47 dB shielding deficit; AgNW susceptible to oxidation degradation over time	HIGH
YInMn Blue spectral coating	UV band protection gap; accelerated UV-induced substrate polymer degradation	MEDIUM
Solid-state cooling	8-13x COP deficit; larger off-grid battery required; food preservation window shorter	HIGH
Microwave RF source	Energy efficiency gap unquantified; magnetron HVT remains GIC failure point	MEDIUM
BFRP rebar design standard	Non-compliant with SDC D/E/F seismic zones; limits geographic deployment	HIGH
Concrete binder	505 vs 113 kg CO ₂ /m ³ ; sustainability mandate unmet; lower strength ceiling	MEDIUM
Aerogel thermal break	33% excess thermal conductivity; 600°C ceiling insufficient for extreme ATE scenarios	MEDIUM
Piezoelectric material	Lead restricted under RoHS Annex II; T _c 117°C lower = premature depolarization risk	HIGH

Parameter Changed	Consequence of Non-Upgrade	Severity
MR fluid specification	9.6% shielding deficit in active damping performance at design operating point	MEDIUM
Indium recovery — Phoenix Protocol	5.4% In lost per cycle; kerosene waste creates RCRA hazardous classification	LOW
Post-event navigation	GPS invalid assumption in Carrington scenario; navigation fails in primary use case	HIGH
Schumann Resonance usage	Pre-event warning capability missed; infrastructure already present but unutilized	LOW
Vibration exposure standards	Compliance claims on obsolete standards; regulatory liability in product disputes	HIGH
Composite matrix — recyclability	No end-of-life recovery path; contradicts Phoenix Protocol; Phoenix Protocol breach	HIGH

Unchanged Elements

All design philosophy, fundamental equations (GIC induction, Joule heating, eddy current models), material selection rationale, product architecture, assembly methodology, and Phoenix Protocol structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | v7 Aegis Home System Draft 7 V1.0 archived; V2.0 is the active specification as of June 2026.